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Project Proposal 04 — Facebook Ego Networks (SNAP, Leskovec–McAuley)

Analisi e Visualizzazione delle Reti Complesse — Network Science module, final project.

Abstract

The Facebook Ego Networks dataset of Leskovec and McAuley (2012) is the union of **ten egocentric friend graphs**. Each is centred on a consenting Facebook user and includes all their friends plus the friendships among those friends. The combined graph has **4 039 nodes** and **88 234 edges**.

The dataset has two features that make it unusually rich:

1. **Anonymised feature vectors** on every node: education, work, hometown, political or religious affiliation, gender, language.
2. **Hand-defined "circles"**: the social groups each user manually drew around different parts of their friend list. These are one of the rare cases of *overlapping* ground-truth communities on a public dataset.

The project uses all this to answer three questions:

1. How strong is homophily on real social attributes?
2. How well do modern community-detection algorithms recover the user-defined circles?
3. How much of the observed structure is already explained by triadic closure?

1. Introduction and motivation

No dataset in this folder exercises more of the syllabus in a medium-scale setting than `ego-Facebook`:

- Its anonymised feature vectors are the natural substrate for **Lecture 5**'s homophily analysis, because the attributes are self-reported rather than inferred.
- Its egocentric construction exposes triadic closure, local bridges, and structural holes — the subject of **Lecture 4** — immediately and visibly.
- Its ground-truth circles are among the very few publicly available overlapping partitions on a real social graph, which makes this the best dataset for stress-testing community-detection algorithms against a criterion that users themselves produced.

Choosing this project means taking the **broadest menu**. Almost every lecture finds a concrete application here, and the ego/union dichotomy lets you move between "what does one neighbourhood look like" and "what does the combined graph of ten neighbourhoods look like". The challenge is the opposite of Proposal 01: here you have to choose *not* to do everything. The temptation is to run every algorithm in the syllabus and produce a fragmented report. **Pick three research questions and treat each properly.**

One useful reference point: McAuley and Leskovec's own circle-detection algorithm, purpose-built for this dataset, reaches **F1 ≈ 0.59** against the ground-truth circles on Facebook. This is a reference ceiling for community-detection

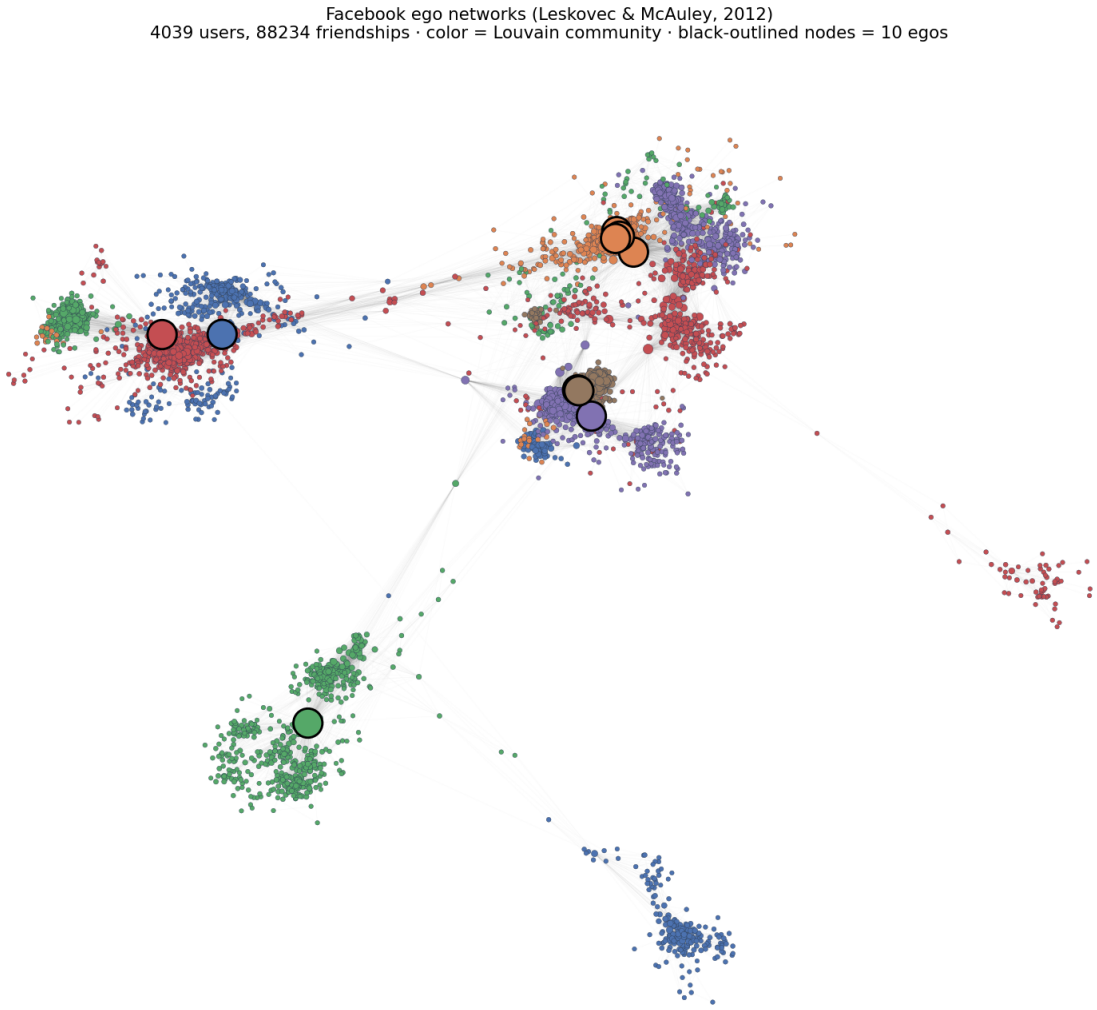

Figure 1: Hero figure: union of ten Facebook ego networks, coloured by Louvain community; the ten egos are drawn larger and black-outlined.

quality on this benchmark. Generic algorithms from the course (Louvain, Leiden, label propagation) typically fall well below it — which is itself a discussion point.

2. The dataset

2.1 Source and provenance

Released with McAuley, J. and Leskovec, J., *Learning to Discover Social Circles in Ego Networks* (NIPS, 2012). Hosted on SNAP at https://snap.stanford.edu/data/ego-Facebook.html. Two archives are provided:

- `facebook_combined.txt.gz` — the merged edge list.
- `facebook.tar.gz` — per-ego feature vectors, ground-truth circles, and the ego's own features. File formats are documented on the SNAP page.

2.2 Structure and semantics

- **4 039 nodes**, **88 234 undirected, unweighted edges**.
- Each node is a Facebook user; each edge is a mutual friendship.
- Per-ego feature names are anonymised hashes (e.g. `education;school;id;anonymized feature 42`). They are structurally usable for assortativity but not semantically interpretable.
- Ground-truth circles are *overlapping*: a friend can belong to several circles, reflecting social roles such as "college friends", "work colleagues", "family".

2.3 Caveats

- The combined graph is the union of ten ego networks, **not** an i.i.d. sample of Facebook. Do not generalise findings beyond these ten neighbourhoods.
- Features are anonymised hashes. Per-feature assortativity is well-defined, but semantic labels are unknown.
- Ground-truth circles are overlapping. They must be evaluated per ego with overlap-aware metrics (overlapping NMI, per-circle F1), not on the combined graph.

3. Research questions

The project should commit to **three or four** of the following. The questions range across most of the syllabus — resist the temptation to take them all. Each uses the ten-ego dataset to answer a specific theoretical claim the course has introduced.

1. **What does a typical ego neighbourhood look like?** An ego network — a user and all their friends, with the friendships among those friends — is a small social system that should, according to the course, look small-world: high clustering (friends of a friend are often friends of each other) and short average paths. **Lectures 4 and 8** argue that such a neighbourhood cannot be modelled by an Erdős–Rényi graph (clustering too high) or by a pure lattice (paths too short) — it should occupy the small-world middle ground. The ten egos in this dataset are an unusually clean sample of real egocentric neighbourhoods. Placing each of them on the small-world plane turns the theoretical claim into an empirical test. The ten points are unlikely to lie on a single mode — some egos will look more Watts–Strogatz-like, others closer to Erdős–Rényi — and that variation across people is itself interesting. *Tools:* density, clustering, and average shortest path per ego; matched Erdős–Rényi and Watts–Strogatz baselines (**Lecture 8**); scatter of `(clustering, avg-path-length)` with all ten egos plus the nulls.

2. **Does Facebook friendship exhibit homophily on real attributes?** Most datasets can only demonstrate homophily with constructed or inferred labels. `ego-Facebook` is different: its feature vectors were self-reported by the users. Per-feature assortativity is therefore a direct measurement of how strongly friendship concentrates on attribute similarity. The question is both empirical ("how strong is the effect?") and comparative ("which features segregate the graph most?"). Because feature names are anonymised hashes, you cannot claim "friendship is most assortative on education" with semantic confidence. But you *can* produce a ranked distribution of assortativity across all features and discuss the shape of that distribution. It is typically

heavy-tailed, with a handful of very strong drivers and a long tail of weaker ones. *Tools:* attribute assortativity coefficient per anonymised feature (**Lecture 5**); ranked table of the top-k; mixing matrices for the most discriminative features.

3. **How much of the observed structure is explained by triadic closure?** Triadic closure — the tendency for two people with a common friend to become friends — is the mechanism that **Lecture 4** identifies as the engine of high clustering in social networks. Granovetter's companion claim is that the edges which do *not* close triangles — the local bridges — are the structurally important ones, because they connect otherwise disconnected parts of the graph. `ego-Facebook` is the right dataset for examining this empirically:

   - The distribution of neighbourhood overlap per edge is typically sharply **bimodal** — most edges embedded in dense triangles, and a small minority being local bridges.
   - The location of those local bridges — particularly whether they are the inter-ego ties — is a striking structural finding.
   - Burt's structural-hole measures (**Lecture 4**) quantify brokerage at the node level and identify which users occupy bridge positions.

   Tools: neighbourhood overlap per edge; local-bridge count (overlap = 0); Burt's constraint and effective size; qualitative inspection of where the local bridges sit.

4. **Is the friendship paradox visible on real data?** Feld's 1991 result — that on average your friends have more friends than you do — is one of the more counterintuitive facts the course covers in **Lecture 7**. It is a direct consequence of degree heterogeneity, but on a real friendship graph it can be felt concretely. On this dataset the **mean degree is around 43** and the **mean degree of a random friend of a random node is well above 100**. The scatter of per-node degree against mean-neighbour-degree shows almost every point above the diagonal, which is the paradox in picture form. *Tools:* mean degree and mean degree-of-neighbours; node-level scatter; the quantitative gap as a summary statistic.

5. **Which centrality measures capture importance in an egocentric graph?** The five centralities of **Lecture 6** plus PageRank (**Lecture 19**) are the course's way of operationalising "importance". The standard exercise is to correlate them and identify disagreements. On an egocentric graph there is a twist: by construction, the ego is connected to every other node in its network, so every centrality measure ranks it first. The *interesting* question is which non-ego node comes **second** under each measure, and whether the second-place node agrees across measures:

   - A stable second-place node across all centralities indicates a well-defined secondary hub.
   - Volatility across measures indicates that "importance" in this ego's world is genuinely multi-dimensional.

   Tools: the five centralities of **Lecture 6** plus PageRank; Spearman rank correlations; per-ego identification of the secondary hubs.

6. **Can modern community-detection algorithms recover the ego-defined circles?** The ground-truth circles are the most interesting feature of this dataset. They are not an algorithmic partition an analyst constructed from attributes — they are *the circles the user themselves drew* around different groups of friends. Whether Louvain, Leiden, Infomap, or label propagation recover these user-defined groupings is therefore the dataset's central test, and it is precisely the evaluation-with-ground-truth question of **Lecture 15**.

   There is a catch the report should address head-on: algorithms produce **hard partitions**, but circles **overlap** (a friend can be both a college classmate and a work colleague). Evaluating a hard partition against overlapping ground truth with a partition metric like NMI underestimates algorithmic quality. Overlapping NMI or per-circle F1 is the fair comparison and yields higher numbers — a methodological point worth making explicitly. *Tools:* Louvain, Leiden, Infomap, label propagation per ego (**Lectures 13–14**); NMI, AMI, overlapping NMI, per-circle F1 against the circles (**Lecture 15**).

7. *(optional)* **How does information cascade through a real friendship network?** Linear-threshold and independent-cascade (**Lecture 18**) are toy models of influence, but this dataset offers a clean substrate on which to run them — a real friendship graph with known structure. Seeding the cascade at the ego, at a random non-hub, and at a structural-hole broker produces three different outbreak curves. The gap between them is the quantitative translation of "centrality matters for spreading". *Tools:* linear-threshold or independent-cascade simulation; final outbreak size as a function of seed centrality.

8. *(optional)* **Can the syllabus's link-prediction features recover withheld friendships?** The traditional-ML pipeline of **Lecture 20** — common neighbours, Jaccard, Adamic–Adar, preferential-attachment, Katz-index — applies naturally to link prediction on a social graph. Withholding 10% of edges and scoring the withheld pairs against held-out negatives produces an AUC per feature, making concrete which structural heuristics are empirically useful. This is a clean standalone experiment and a strong extension to the core structural analysis. *Tools:* train/test edge split; feature computation per candidate pair; AUC evaluation.

4. Expected findings

A well-executed project should reach, at minimum, the following results:

- Per-ego clustering in the **0.35–0.60 range** and average path length around **3–4** — well above Erdős–Rényi expectation. The ten egos span a spectrum, some close to Watts–Strogatz, others more sparsely interconnected.
- A ranked distribution of attribute assortativity in which the top-3 features exceed **+0.6** and the top-20 are all positive. Reporting the full distribution across all anonymised features is itself a finding.
- A strongly bimodal neighbourhood-overlap distribution: most edges embedded in dense triangles (overlap > 0.2), a minority being local bridges. Local-bridge fraction typically **3–7%** of edges.
- Friendship paradox easily demonstrable: mean degree ≈ **43**, mean degree-of-neighbours ≈ **100+** — a factor of roughly **2.5**.
- Community-vs-circles NMI around **0.35–0.55** per ego on Louvain/Leiden — lower than the apparent visual separation of the ego-network suggests, because of the hard-partition vs overlapping-circles mismatch. Overlapping-NMI or per-circle F1 is typically higher and is the fair comparison. As a reference, McAuley and Leskovec's own specialised circle-detection method reported **F1 ≈ 0.59** on the Facebook data in their 2012 NIPS paper — a modest-looking number that shows how hard the problem is even with a purpose-built algorithm.
- Observed clustering roughly **an order of magnitude above** the matched ER null.

5. Visualization plan

Visual design is the group's responsibility — the dataviz module covers the *how*. This section only states the *what*: for each finding the project claims, a visualisation should carry the corresponding message.

- **The shape of the union.** The report's *hero figure*: a view of the combined graph in which the ten egocentric neighbourhoods are visible simultaneously, with the bridges that tie them together.
- **"How assortative is each feature?"** A visualisation that ranks the anonymised features by assortativity and exposes the distribution of homophily strengths.
- **"Are ties strong, weak, or bridging?"** A visualisation of the neighbourhood-overlap distribution with local bridges highlighted.
- **"Does the friendship paradox hold?"** A visualisation comparing per-node degree with mean neighbour degree across the population.
- **"Which community-detection algorithm best recovers the circles?"** A visualisation, per ego, comparing algorithmic partitions against the ground-truth circles across multiple quality metrics.
- **"What does one ego network look like in detail?"** A focused figure on one selected ego, enriched with feature and circle information, that lets the reader understand a single neighbourhood well.

An interactive view of a single ego's network — showing features and circle memberships on hover, and letting the reader toggle between algorithmic community and ground-truth circle colouring — is optional but strongly rewarded.

6. Possible extensions

- **Per-ego comparative study.** Are some egos structurally "broker-like" and others "tightly clustered"? A 2-D phenotype space of `(clustering, avg-path-length)` per ego reveals the spectrum.
- **Weighted variant.** Use feature-vector similarity as edge weight and ask whether weighted community detection improves the match against ground-truth circles.
- **Link-prediction comparison.** Extend question 8 by comparing traditional feature-based scores with a node2vec embedding (**Lecture 21**) on the same held-out edges.

- **Robustness study.** Targeted versus random node removal — are the egos themselves the critical nodes?

7. Common pitfalls

- **Treating the ego as "just another node" is the classic mistake.** By construction the ego connects to every other node in its network and ranks first under every centrality, so any honest table either explains this or drops the ego.
- **Evaluating community detection with a partition metric (NMI, ARI) on overlapping ground truth** underestimates the algorithms' quality. Use overlap-aware metrics.
- **Hashed features invite over-interpretation.** Be explicit that semantic labels are unknown.
- **Stochastic algorithms require seeded runs and stability analysis.**

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- SNAP project page: https://snap.stanford.edu/data/ego-Facebook.html.
- Course syllabus and lecture slides: `netsci/ns_syllabus.md`, `netsci/pdfs/`.